

Customer Churn Analysis

Data Cleaning & Structural Validation — Completed Submission

Objective

Transform the supplied raw customer churn data into a reliable, analysis-ready format using Python/Pandas and Excel.

Cleaning Results

Check	Result
Rows inspected	15
Duplicate rows removed	0
Missing cells found	0
Rows after cleaning	15
Column headers standardized	Yes
Data types validated	Yes
Categorical/text values standardized	Yes
Cleaned CSV exported	Yes

Tasks Completed

- Inspected the dataset for missing values, duplicates, and data-type mismatches.
- Removed duplicate records where present.
- Checked numeric fields and converted them to numeric types.
- Standardized column names, categorical strings, and contract labels.
- Exported the cleaned dataset to CSV and Excel.
- Created a reproducible Jupyter Notebook containing the cleaning steps.

Deliverables

1. cleaned_customer_churn.csv — cleaned dataset
2. customer_churn_cleaned.xlsx — cleaned Excel workbook with summary
3. customer_churn_cleaning.ipynb — Jupyter Notebook with Python/Pandas workflow

Important: The dataset values were transcribed from the customer-churn screenshot supplied in this chat. If your portal's downloadable sample contains additional rows or different values, upload that original file and I can regenerate the deliverables from the exact source.